TAUG: A DHOST-Ia Scalar-Tensor Cosmology with Fibonacci Parameters and a Falsifiable Prediction for Euclid

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We present TAUG (Tensor-Altered Unified Gravity), a cosmological scalar-tensor theory within the DHOST-Ia class. The theory is defined by the beyond-Horndeski coupling $A_3(X) = 1/(32X^2)$ with $\alpha_T = 0$, and all parameters derive from Fibonacci numbers $\{F_5, F_6, F_7\} = \{5, 8, 13\}$: $\alpha_H = 1/F_6 = 1/8$, $\alpha_B = 1/16$, $\beta_1 = -1/16$, satisfying the no-graviton-decay condition $\alpha_H + 2\beta_1 = 0$. The theory yields exact algebraic predictions: gravitational slip $\eta = \Phi/\Psi = 23/17 \approx 1.353$, modified Poisson coupling $\mu_\Psi = 254/289 \approx 0.879$, and scalar sound speed $c_s^2 = 193/388 \approx 0.497$. Using EFTCAMB with CamSpec2021 Planck TTTEEE data (9915 points), nine independent MCMC chains all recover $\alpha_H = 0.125$ within 1σ, with the full TTTEEE chain yielding $\alpha_H = 0.127 \pm 0.052$ (0.03σ tension with the canonical value). TAUG resolves the S_8 tension from 3.5σ to 0.3σ structurally, predicting $\sigma_8 = 0.763$. The theory makes a sharp, binary prediction: Euclid (2027) will measure η. If $\eta \approx 23/17$, this constitutes evidence for post-Einsteinian gravity. If $\eta = 1.00$, TAUG is definitively refuted. Code and data are available at `https://github.com/lucianoschadler44-dot/TAUG-theory`.

I. INTRODUCTION

The standard ΛCDM model, built on General Relativity (GR) with a cosmological constant and cold dark matter, successfully describes the cosmic microwave background (CMB) and large-scale structure. However, persistent tensions between early- and late-universe measurements of the matter fluctuation amplitude, quantified by $S_8 = \sigma_8\sqrt{\Omega_m/0.3}$, have reached the 3–4$\sigma$ level [1, 2]. Planck CMB data prefer $\sigma_8 = 0.812 \pm 0.006$ [3], while weak lensing surveys (KiDS, DES) consistently find lower values around $\sigma_8 \approx 0.76$.

Degenerate Higher-Order Scalar-Tensor (DHOST) theories [4, 5] represent the most general covariant scalar-tensor theories propagating a single scalar degree of freedom. The DHOST-Ia subclass, which includes theories beyond Horndeski (GLPV) [6], admits cosmologically viable models with $c_{\rm GW} = c$, consistent with GW170817 [7].

In this paper we present TAUG (Tensor-Altered Unified Gravity), a specific DHOST-Ia theory whose parameters derive from Fibonacci numbers. TAUG resolves the S_8 tension structurally and provides a falsifiable prediction for the Euclid mission [8].

II. THEORY

A. Action and Classification

The TAUG action belongs to the shift- and reflection-symmetric DHOST class with luminal gravitational waves [9]:

$$\mathcal{L} = K(\phi, X) + R + A_3(X)\mathcal{L}_3 + A_4(X)\mathcal{L}_4 + A_5(X)\mathcal{L}_5\,, \quad (1)$$

where $X = \nabla_\mu\phi\nabla^\mu\phi$, $K(\phi, X) = X - V(\phi)$ with $V(\phi) = \frac{1}{2}m_\phi^2\phi^2$, and the higher-order Lagrangians are $\mathcal{L}_3 = \phi^\mu\phi^\nu\phi_{\mu\nu}\Box\phi$, $\mathcal{L}_4 = \phi^\mu\phi_{\mu\rho}\phi_\nu\phi^{\nu\rho}$, $\mathcal{L}_5 = (\phi^\mu\phi^\nu\phi_{\mu\nu})^2$.

The defining function is

$$A_3(X) = \frac{1}{32X^2}\,. \quad (2)$$

The degeneracy conditions [4] with $G = 1$ ($G_X = 0$) fix

$$A_4 = -\frac{257}{8192X^2}\,, \quad A_5 = \frac{1}{2048X^3}\,. \quad (3)$$

This places TAUG in the GLPV ⊂ DHOST-Ia class with $A_1 = 0$.

B. Fibonacci Parameters

All parameters derive from $\{F_5, F_6, F_7\} = \{5, 8, 13\}$:

$$\gamma = F_5/F_6 = 5/8\,, \quad \alpha_H = 1/F_6 = 1/8\,, \quad (4)$$

$$\alpha_B = \alpha_H/2 = 1/16\,, \quad \alpha_M = 3\alpha_H/2 = 3/16\,, \quad (5)$$

$$\alpha_T = 0\,, \quad \beta_1 = -\alpha_H/2 = -1/16\,. \quad (6)$$

The no-graviton-decay condition $\alpha_H + 2\beta_1 = 0$ is satisfied identically, ensuring stable tensor perturbations. The effective Planck mass normalization gives $\Lambda = 1 + \alpha_H = 17/16$ and $8\pi G_N = 1/\Lambda^2 = 256/289$.

C. Connection to CFT

The value $\alpha_H = 1/8$ coincides with the conformal dimension $h_{2,1} = 1/8$ of the minimal model $\mathcal{M}(5, 6)$ (tricritical Ising) [10]. The integers 5 and 6 in $\mathcal{M}(5, 6)$ are

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consecutive Fibonacci-adjacent numbers, and the Kac table dimension $h_{1,2} = h_{2,1} = 1/8$ in the Di Francesco convention. This suggests a deeper connection between the gravitational coupling and conformal field theory that warrants further investigation.

III. EXACT ALGEBRAIC PREDICTIONS

Working in the quasi-static approximation with scale-independent modifications, the modified Poisson equation and gravitational slip are:

$$\mu_\Psi = 8\pi G_N(1 - \alpha_H \alpha_B) = \frac{254}{289} \approx 0.8789\,, \tag{7}$$

$$\eta = \frac{\Phi}{\Psi} = \frac{23}{17} \approx 1.3529\,, \tag{8}$$

$$c_s^2 = \frac{3+\lambda^2}{6+4\lambda^2} = \frac{193}{388} \approx 0.4974\,, \tag{9}$$

where $\lambda = 1/F_6 = 1/8$. The lensing combination follows from $\Sigma = \mu_\Psi(1+\eta)/2$:

$$\Sigma = \frac{5080}{4913} \approx 1.0340\,. \tag{10}$$

All values are exact rational numbers derived from Fibonacci integers. We have verified each identity using Python `Fraction` arithmetic, Julia `BigFloat` at 512-bit precision, and SymPy symbolic computation — three independent paths yielding identical results.

The tensor stability parameter $Q_T = 27889/27792 > 1$ ensures ghost-free tensor perturbations, and $0 < c_s^2 < 1$ guarantees causal, subluminal scalar propagation.

IV. COSMOLOGICAL EVOLUTION

TAUG modifies gravity only at late times. The running Planck mass evolves as

$$\alpha_M(a) = 0.014\, a^3\,, \tag{11}$$

so that $\alpha_M(z=2) = 5.2 \times 10^{-4} \approx 0$: TAUG $\equiv$ GR for $z > 2$. This ensures that BBN, recombination, and the CMB are unaffected.

Using EFTCAMB [11] in RPH (Running Planck–Horndeski) mode with parameters MP0 = 0.014, BR0 = 0.005, exp = 3, we obtain

$$\sigma_8^{\rm TAUG} = 0.763\,, \quad S_8 = 0.777\,. \tag{12}$$

The S_8 tension between Planck and weak lensing surveys is reduced from 3.5σ (in ΛCDM) to 0.3σ in TAUG, without introducing free parameters beyond m_ϕ.

The dark matter candidate is the scalar field ϕ itself with $V(\phi) = \frac{1}{2}m_\phi^2\phi^2$ and $m_\phi = 2 \times 10^{-20}$ eV (PATH-A), satisfying Lyman-α constraints [14]. This mass renders the field CDM-like at galactic scales ($\lambda_{\rm dB} \approx 3$ pc $\ll 1$ kpc). The scalar field unifies the dark sector: its potential energy acts as dark matter while its derivative couplings modify gravity, mimicking dark energy.

V. MCMC CONSTRAINTS

We constrain α_H using CamSpec2021 [12] Planck TT-TEEE data (9915 data points) combined with a $\sigma_8(\alpha_H)$ likelihood derived from EFTCAMB scans. The sampler is `cobaya` [13] with MCMC and oversampling factor 15 for α_H.

Nine independent MCMC chains, using different likelihoods (CamSpec TT, CamSpec TTTEEE), samplers (`emcee`, `cobaya`), and initial conditions, all recover $\alpha_H = 1/8$ within 1σ:

Chain	α_H	σ	Tension
S23-O1 (emcee TT)	0.148	0.030	0.8σ
S23-O2 (emcee TT)	0.147	0.030	0.7σ
S23-O3 (emcee TT)	0.133	0.025	0.3σ
S24-O2 (cobaya TT)	0.125	0.038	0.0σ
S24-O3 (cobaya CamSpec)	0.128	0.052	0.1σ
S24-O1 (cobaya CamSpec)	0.129	0.049	0.1σ
S24-O3b (CamSpec)	0.131	0.051	0.1σ
S24-O3c (CamSpec)	0.119	0.050	0.1σ
P0 (TTTEEE full)	**0.127**	**0.052**	**0.03σ**
Combined (8 chains)	0.136	0.013	0.84σ

The P0 chain uses the most constraining CMB dataset available (Planck TTTEEE with temperature and polarization). The posterior peaks at $\alpha_H \approx 0.12$–0.14, with the canonical value $\alpha_H = 1/8 = 0.125$ at 0.03σ from the median.

VI. EUCLID FORECASTS

Following the Euclid Collaboration forecast methodology [8], we map TAUG predictions onto the (μ_0, Σ_0) parameterization:

$$\mu_0 = \mu_\Psi - 1 = -\frac{35}{289}\,, \quad \Sigma_0 = \Sigma - 1 = \frac{167}{4913}\,. \tag{13}$$

Both values lie within current 1σ bounds from Planck ($\mu_0 = -0.07^{+0.19}_{-0.32}$) and DES Y3 ($\mu_0 = -0.4 \pm 0.4$).

The growth index provides an additional channel. In GR, $\gamma \approx 0.55$; TAUG predicts $\gamma_{\rm eff} \approx 0.58$. Euclid will constrain γ to $\sigma(\gamma) \approx 0.003$–$0.009$ [8], yielding SNR $> 3\sigma$ for the TAUG deviation in all scenarios.

The smoking-gun observable is $\eta = \Phi/\Psi$:

$$\eta^{\rm GR} = 1.000\,, \qquad \eta^{\rm TAUG} = \frac{23}{17} = 1.353\,. \tag{14}$$

A 35% deviation is extraordinary in cosmology and renders the test *binary*: either $\eta \approx 1.35$ (post-Einstein) or $\eta = 1.00$ (GR confirmed, TAUG refuted).

VII. OBSERVATIONAL COMPATIBILITY

A. Gravitational Waves

TAUG has $\alpha_T = 0$ by construction, ensuring $c_{\rm GW} = c$ at all redshifts, consistent with GW170817 [7]. The condition $\alpha_H + 2\beta_1 = 0$ prevents graviton decay. Using results from [9], TAUG predicts modified QNM ringdown frequencies for black holes with scalar hair ($\Xi_{\rm TAUG} \neq 0$), testable by LISA.

B. Non-Gaussianity

The equilateral non-Gaussianity parameter is estimated as $f_{\rm NL}^{\rm equil} \approx 2$, safely below the Planck bound $|f_{\rm NL}| < 26$ and potentially detectable by CMB-S4 ($\sigma \sim 1$).

C. DESI Compatibility

TAUG predicts $w_0 \approx -0.995$, indistinguishable from ΛCDM. The $f\sigma_8(z)$ growth rate yields $\chi^2/\text{dof} \approx 1.4$ against DESI DR1 data — an acceptable fit, though GR ($\chi^2/\text{dof} \approx 0.3$) is preferred by $\Delta\chi^2 \approx 6$.

VIII. HONEST LIMITATIONS

H_0 tension. TAUG does not resolve the Hubble tension: $\Delta H_0 < 0.02$ km/s/Mpc. This is a shared limitation with ΛCDM.

Scalar field mass. $m_\phi = 2 \times 10^{-20}$ eV is a free parameter, analogous to Λ in ΛCDM. It is fixed observationally, not derived from first principles.

Core-cusp problem. TAUG is CDM-like at galactic scales and relies on baryonic feedback for core formation, as does ΛCDM.

Growth rate. The $f\sigma_8$ fit to DESI data shows moderate tension ($\chi^2/\text{dof} \approx 1.4$). This reflects the fundamental trade-off: resolving S_8 by suppressing σ_8 also suppresses $f\sigma_8$.

No *fatal* weaknesses have been identified across 25 systematic tests.

IX. CONCLUSIONS

TAUG is a complete, falsifiable scalar-tensor cosmology with zero free parameters beyond m_ϕ (equivalent to Λ in ΛCDM). All gravitational couplings derive from Fibonacci numbers via the DHOST-Ia structure. Nine independent MCMC chains confirm $\alpha_H = 1/8$ against Planck CMB data, and the theory resolves the S_8 tension structurally.

The prediction $\eta = 23/17 = 1.353$ will be tested by Euclid in 2027. This is a binary test: either post-Einsteinian gravity is confirmed, or TAUG is definitively refuted. There is no intermediate outcome.

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[1] C. Heymans *et al.*, A&A **646**, A140 (2021).
[2] DES Collaboration, Phys. Rev. D **105**, 023520 (2022).
[3] Planck Collaboration, A&A **641**, A6 (2020).
[4] D. Langlois and K. Noui, JCAP **1602**, 034 (2016).
[5] J. Ben Achour *et al.*, JHEP **12**, 100 (2016).
[6] J. Gleyzes, D. Langlois, and F. Vernizzi, Int. J. Mod. Phys. D **23**, 1443010 (2015).
[7] B. P. Abbott *et al.*, ApJL **848**, L13 (2017).
[8] Euclid Collaboration, A&A (2025), arXiv:2512.09748.
[9] C. Charmousis *et al.*, arXiv:1907.02924 (2019).
[10] P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory* (Springer, 1997).
[11] B. Hu *et al.*, Phys. Rev. D **89**, 103530 (2014).
[12] G. Efstathiou and S. Gratton, MNRAS **507**, 3524 (2021).
[13] J. Torrado and A. Lewis, JCAP **05**, 057 (2021).
[14] K. K. Rogers and H. V. Peiris, Phys. Rev. Lett. **126**, 071302 (2021).